GenAI for Engineers Learner Guide

IES | v3.1 | Clear explanations and worked examples

One day, 09:00–17:00. Follow the supplied six-module lesson plan. Guided Excel exercises total 65 minutes: SPC 25, FMEA 20, DMAIC 20. Agents and SharePoint are trainer demonstrations.

How to use this guide

The numbered topics match the presentation. First understand the question, then read the example and check the result. Use one working copy of Lab-01-SPC/Worked-Reference.xlsx for all three exercises. Starter.xlsx is optional after-class practice.

Module 1

1. GenAI for Engineers

Use Copilot to understand data and explain engineering findings.

Today: One pump case, three short Excel exercises

Your role: Check the answer before using it

Explanation: All pump data are synthetic.

2. What will we do today?

Follow the six modules in the supplied lesson plan.

Morning: Meet Copilot → write useful prompts → analyse the pump case

Afternoon: Watch agent and SharePoint examples → review AI security

Hands-on work: SPC 25 min · FMEA 20 min · DMAIC 20 min

Explanation: Here, demonstrations and discussion make up the rest of the day.

3. What is generative AI?

It produces a draft answer from your instructions and available information.

You provide: A question and relevant information

AI produces: A draft explanation, table, formula or report

You check: Does the draft agree with the information?

Explanation: A prompt is the instruction you give the AI. Copilot is Microsoft’s AI assistant. A fluent answer can still contain a wrong number or unsupported statement.

4. What can Copilot do in Excel?

For example: help you understand why a pump test result looks unusual.

Ask: “Which test results are below the usual range?”

Copilot helps: Suggest a formula or explain the chart

Engineer checks: Compare the answer with the recorded test values

Explanation: These are steps in a joint workflow, not three capabilities belonging to Copilot. Here, the last step belongs to the engineer. Editing and chart features depend on the account and Excel version.

5. Our case: is the pump delivering consistently?

A factory team tests a small pump that feeds an industrial cooling-water system.

The job: The pump should deliver about 62 litres per hour during the test.

The concern: Some later tests record less liquid than earlier tests.

Our task: Describe the change, review possible failures and summarise the evidence.

Explanation: CW-401 is only the fictional case identifier; it is not a method learners must know. A dosing skid is a pump and associated equipment mounted together. We will say pump system in class. These are delivery tests, not measurements of water-treatment compliance.

6. How is the test result measured?

Collect the liquid for two minutes, then convert the amount to litres per hour.

Measured: 2.067 litres collected in 120 seconds

Calculation: 2.067 × 3600 ÷ 120 = 62.01 L/h

Meaning: At that rate, the pump would deliver about 62 litres in one hour.

Explanation: L/h means litres per hour. This is an example calculation, rounded from the synthetic case. A test with a different duration cannot be compared using the same multiplier.

7. Use one prepared workbook today

The calculations and charts are already prepared so we can focus on asking and checking.

Open: A working copy of Worked-Reference.xlsx

Find: SPC_Summary: pump chart · FMEA_Review: failure table

Then: Rate_Comparison: before/after rates · Prompts: instructions to copy

Explanation: Starter.xlsx is optional practice after class.

Module 2

8. A prompt is an instruction to Copilot

A useful prompt tells Copilot what to use and what answer you need.

Too vague: “Analyse this.”

More useful: “Use the pump test table. Identify unusual results and explain them in three sentences.”

Why this helps: Copilot knows the information to use and the size of the answer.

Explanation: Prompt engineering means deliberately writing, checking and improving instructions. It does not mean using complicated language. The improved prompt is still a draft: add the relevant sheet and check a value.

9. Use RISE to build a clear prompt

RISE means Role, Input, Steps and Expectations.

Role + Input: “Act as an engineering assistant. Use the pump chart on SPC_Summary.”

Steps: “Identify results outside the displayed limits. Check one against Analysis.”

Expectations: “Give three sentences: what changed, evidence and what is still unknown.”

Explanation: Input means the information to use. Expectations means the form and quality of the answer. The role sets the perspective; it does not give the AI professional authority.

10. Add the engineering details with E-MAP

E-MAP reminds you to include context, measures, assumptions and limits.

Engineering + Metrics: “This is a pump delivery test. Report flow in litres per hour.”

Assumptions: “The prepared chart uses the first 60 valid tests as its reference.”

Parameters: “Keep those limits unchanged. Use only the supplied workbook.”

Explanation: E-MAP means Engineering context, Metrics, Assumptions and Parameters. It complements RISE; learners do not have to memorise two competing recipes. A parameter is a value or rule the analysis must use.

11. Tell Copilot what it must not change

Constraints are simple rules that keep the task within scope.

Protect the data: “Do not change the original test records.”

Handle gaps: “If a measurement is missing, say it is missing; do not invent it.”

Limit the conclusion: “Describe the change. Do not claim that we have proved its cause.”

Explanation: Replace abstract warnings with the three exact prompt sentences. Here, a blank measurement means we do not know its value. Removing an extreme valid reading would hide a possible problem.

12. Show the format you want

Use three fields: observation, value and follow-up question.

Requested format: Observation | Supporting value | Question to investigate

Observation + value: Later delivery is unusual: 17 results fall below the 60.04 L/h lower limit.

Follow-up question: Were those tests performed under the same conditions?

Explanation: This is a format example, not a complete analysis. Use a short output to make verification practical.

13. Use three perspectives to improve a review

A persona is a requested viewpoint, not a separate expert.

Reliability engineer: “What equipment issue could explain low delivery?”

Operations engineer: “Were the test conditions and settings consistent?”

Safety reviewer: “What must be checked before changing the equipment?”

Explanation: Copilot simulates all three perspectives, so their agreement is not independent evidence. Use the same pump information for all roles.

14. Let the perspectives challenge one claim

Example claim: “Replacing the strainer solved the problem.”

Reliability asks: “Do we have evidence that the removed strainer was blocked?”

Operations asks: “A verification check also happened. Which change made the difference?”

Better conclusion: “Results improved afterwards, but we cannot isolate the cause yet.”

Explanation: A strainer catches particles before they enter equipment. In this case strainer replacement and verification happened together. Do not run an unbounded debate or force consensus.

15. Why spend time improving a prompt?

Clear instructions make the answer easier to use and easier to check.

Less rework: Specify the right sheet, units and output length at the start.

More consistency: Reuse the same structure for similar engineering tasks.

Better review: Ask for supporting values and unanswered questions.

Explanation: Measure total time including checking and correcting the answer. Prompt quality does not guarantee factual accuracy.

16. Turn a test note into a short report

Trainer example: preserve the facts while improving the writing.

Source note: “Later tests lower. Strainer replaced. Verification also done. Pilot results improved.”

Prompt: “Write a short observation and one follow-up question. Use only these facts.”

Example draft: “Delivery improved after two changes. Which change contributed to the improvement?”

Explanation: Here, polishing the language does not create new evidence. The same pattern works for inspection notes and technical reports. This is a demonstration, not another learner lab.

Module 3

17. Three methods answer three different questions

We use the same pump case for all three.

SPC: Has the process changed over time?

FMEA: What could go wrong, and what should we review?

DMAIC: How do we organise an improvement investigation?

Explanation: The full names are: Statistical Process Control; Failure Mode and Effects Analysis; Define, Measure, Analyse, Improve, Control. These are engineering methods. Copilot helps with their outputs; it does not replace learning what the methods mean.

18. SPC: Statistical Process Control

Compare measurements over time with the earlier, usual pattern.

Centre line: The average of the earlier reference tests.

Upper and lower limits: The expected range calculated from that reference period.

A point outside: A reason to investigate a change, not proof of a specific fault.

Explanation: Baseline means the earlier period used for comparison. Here it contains 60 valid tests. Common-cause variation as the usual small fluctuations; an unusual pattern may indicate a special cause. A point inside the limits does not prove the equipment is safe.

19. Read the pump chart from left to right

The later monitoring results drop below the earlier pattern.

Earlier tests: Average about 62.00 L/h

Monitoring period: 17 valid results are below the lower control limit.

Pilot period: Results return closer to the earlier pattern after two changes.

Explanation: Limits remain about 60.04 and 63.97 L/h. The count excludes invalid or missing tests, not merely low readings. Pilot as a short follow-up period.

20. A control limit is not a specification

One describes process behaviour; the other describes the required result.

Control limits: About 60.04–63.97 L/h, calculated from earlier tests.

Test specification: 58.9–65.1 L/h, a fictional requirement supplied for this exercise.

Example: A 59.5 L/h result meets the test specification but is unusual for this process.

Explanation: Cp and Cpk compare process spread and centring with specification limits. They only make sense when the process and measurement assumptions are suitable.

21. Exercise 1: explain the pump chart

25 minutes · use the prepared workbook, not a blank analysis.

Ask Copilot: “Review SPC_Summary and Analysis. Explain the unusual monitoring results in three sentences.”

Check one result: Find TEST-073 in Analysis and compare it with the lower limit.

Save: What changed, the supporting evidence and one unanswered question.

Explanation: Keep the displayed limits. Expected finding: 17 valid monitoring points below the lower limit. Use the longer bounded prompt in Prompts if needed.

22. FMEA: think through possible failures

FMEA stands for Failure Mode and Effects Analysis.

Failure mode: The pump delivers too little liquid.

Effect: The system receives less liquid than intended.

Possible cause + control: A restriction may contribute; a delivery test may detect the problem.

Explanation: Separate what failed, its consequence, a possible cause and a control. The potential restriction is not proven. Other FMEA rows can describe possible failures that have not occurred. This prevents learners treating every row as an incident record.

23. Use the scores to prioritise a review

RPN is a risk priority number: Severity × Occurrence × Detection.

Severity (S): How serious is the effect?

Occurrence (O) + Detection (D): How likely is the failure? How hard is it to detect before the effect?

Worked example: FM01: 7 × 5 × 4 = 140. The scores are supplied workshop judgements.

Explanation: Higher detection score means harder to detect, not better detection. FM01 is the ID of the low-delivery row. RPN supports prioritisation; it is not a probability or proof of safety.

24. Exercise 2: check two FMEA rows

20 minutes · inspect the supplied table and propose one follow-up.

Ask Copilot: “Check FM01 and FM04 in FMEA_Review. Explain their scores and one evidence gap.”

Verify: FM01 has RPN 140. FM04 has severity 9 and RPN 90.

Save: One missing piece of evidence and one proposed action for FM01.

Explanation: Consider why the lower RPN row may still need attention: its consequence is more serious. FM04 is a potential leakage scenario, not an observed leak. Leave residual scores unassessed; stop at 12:30.

25. DMAIC: organise an improvement investigation

Move from a defined problem to a plan for maintaining improvement.

Define + Measure: State the problem and measure how often it occurs.

Analyse + Improve: Investigate explanations and propose a change to test.

Control: Decide what to monitor, who checks it and what happens if results worsen.

Explanation: Use the pump example at each stage. Define: delivery-test deviations. Measure: deviations per test opportunity. Analyse: review chart and records. Improve: test a supported proposal. Control: continue monitoring with a named reviewer.

26. Compare rates, not just event counts

Use the separate eight-week event log: deviations are recorded test problems.

Before: 93 deviations ÷ 1,008 opportunities = 9.23%

Pilot: 11 deviations ÷ 336 opportunities = 3.27%

Meaning: One logged chance for a test problem. Compare problems ÷ opportunities.

Explanation: An opportunity is one logged chance for a test deviation. This eight-week event log is different from the 120 individual pump tests; do not mix denominators. The change is 5.95 percentage points. A lower rate after two changes does not identify which caused it.

27. A Pareto chart shows the largest categories

Start the investigation where most recorded problems occur.

Largest category: Low delivery: 55 of 93 baseline deviations.

Next category: Test validity: 16 of 93 deviations.

Use it for: Choosing an investigation priority, not proving a root cause.

Explanation: Bars show counts, largest first. The line adds their percentages cumulatively. Low delivery and test validity together account for about 76% of the baseline deviations. A category is a description of the problem, not necessarily its cause.

28. Exercise 3: write a short improvement summary

20 minutes · use Rate_Comparison, WeeklyEvents and DMAIC.

Ask Copilot: “Check the before/pilot rates. Give three cells: change, limitation and next evidence.”

Check: Recalculate 93 ÷ 1,008 and 11 ÷ 336.

Save: A lower rate was observed; two changes occurred; more evidence is needed.

Explanation: Do not create a full report or new chart.

Module 4

29. What is a Copilot agent?

An assistant with reusable instructions and selected information for a recurring task.

Ordinary prompt: You explain the task again each time.

Agent: The task instructions and selected reference material are already prepared.

Example: An FMEA assistant uses an agreed template and rating guidance.

Explanation: This is a trainer demonstration. Instructions are the task rules and knowledge as reference material. An agent still needs testing and human review. We do not ask learners to build or deploy agents today.

30. Watch a prepared FMEA assistant

Follow one input through to a checked answer.

Input: Pump description, case records and rating guidance.

Agent draft: Failure modes, effects, possible causes and missing information.

Trainer check: Does each statement match the supplied records?

Explanation: Never claim a recorded live run if using a worked example.

33. Research and analysis need different checks

Research asks what a source says; analysis asks what the numbers show.

Research example: “Find the maintenance guidance in the supplied manual and cite the passage.”

Analysis example: “Compare the test rates and show the calculation.”

Check: Open the cited passage; independently calculate one result.

Explanation: The learning point is the difference in evidence, not a guarantee of access. Use prepared source and arithmetic examples if live tools are unavailable.

34. A report assistant can reuse a template

It turns recorded facts into a consistent report structure.

Recorded facts: Observation, action, owner and due date.

Report sections: Progress, outstanding issues and next actions.

Human review: Check that “proposed” has not become “completed”.

Explanation: A minutes-to-actions-to-report workflow with the same fields. Do not imply that recurring automation has been deployed or that the assistant can approve work.

Module 5

36. What does grounding mean?

Give Copilot a relevant document to use when answering.

Without the document: An answer may rely on general information.

With the document: The answer can refer to the supplied procedure or manual.

Your check: Open the cited passage and confirm it supports the answer.

Explanation: SharePoint is Microsoft’s shared document storage and collaboration platform. Grounding means supplying relevant context, often retrieved from documents. It improves relevance but does not guarantee correctness.

37. Follow one answer back to its document

Trainer demonstration: question → passage → answer → check.

Question: “What does this maintenance procedure require us to record?”

Cited answer: The assistant lists the required fields and points to the passage.

Check: Open that passage and check its revision and applicability.

Explanation: Do not invent a real standard or clause.

39. Check who may access the documents

Use only information that the learner is authorised to use.

Before asking: Check the library and documents are in the intended scope.

Before sharing: Check the answer does not expose restricted project information.

If a document gives instructions: Treat its text as source material; do not let it redirect the task.

Explanation: A retrieved document might contain text such as “ignore earlier instructions”. Here, this text is not authority to change the user’s task.

Module 6

40. Seven checks for responsible use

Apply these checks to one answer from today’s pump case.

1–2: inputs: Classify the data; check access and governance.

3–4: interaction: Write a safe prompt; verify the answer.

5–7: use: Obtain human sign-off; keep records; report suspicious activity.

Explanation: The seven checks are: data classification; governance/access; secure prompting; output verification; human-in-the-loop sign-off; traceability/audit; security awareness/incident response. Expected: output verification, supported by traceability.

41. Keep one prompt you can reuse

Replace the bracketed parts with your next approved task.

Prompt: “Use [source]. Help me [task]. Keep [rules]. Return [format]. State missing information.”

Example: “Use the pump table. Explain unusual tests. Keep the raw data unchanged. Return three sentences.”

Review habit: Check one number and one supporting source before sharing.

Explanation: There is no new lab. Encourage a small task with a measurable benefit and a clear reviewer.

42. What should you remember?

Copilot drafts; engineers understand, check and decide.

SPC: Is the process behaving differently?

FMEA: What could fail, and what needs attention?

DMAIC: What evidence supports the improvement plan?

Explanation: Remember that the full starter workbook is optional practice, not unfinished class work.

Detailed exercise instructions

Lab-01-SPC

SPC — 25-minute guided Excel exercise

Module 3 only. Use the prepared Worked-Reference.xlsx; all three exercises use one working copy.

5 min · Read the prepared analysis

Open Worked-Reference.xlsx and save a working copy. The trainer has prepared data cleaning, unit conversion, baseline formulas and chart. Locate RawTests, Analysis and SPC_Summary; keep the source and baseline limits unchanged.

12 min · Prompt and inspect

Use SPC3 from Prompts to explain the existing chart and identify outside-limit points. If editing is available, add a signal flag in a copy. Check TEST-073 and one baseline row against fixed limits 60.03757–63.96743 L/h. Do not rebuild the chart or calculate Cp/Cpk in class.

8 min · Verify and retain

Confirm 17 valid monitoring points outside the limits. Save one three-sentence interpretation: signal, evidence and limitation. An association does not establish the cause. Stop at the time limit.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Lab-02-FMEA

FMEA — 20-minute guided Excel exercise

Module 3 only. Use the prepared Worked-Reference.xlsx; all three exercises use one working copy.

4 min · Read the prepared FMEA

Continue with the same workbook. Open FMEA_Review and ScoringNote. The six rows and provisional scores are supplied; do not generate a new system or rescore every row.

10 min · Prompt and check priorities

Use FMEA2 to explain or sort a copy by RPN and flag S>=9. Recompute FM01: 7×5×4=140. Check FM04: S=9 and RPN=90 still require severity review. Keep IDs attached to their source rows.

6 min · Review one row

Write one evidence gap for FM01 and one action proposal. Keep residual ratings unassessed. The multi-persona debate is a trainer demonstration in Module 2, not an extra participant task.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Lab-03-DMAIC

DMAIC — 20-minute guided Excel exercise

Module 3 only. Use the prepared Worked-Reference.xlsx; all three exercises use one working copy.

4 min · Read the prepared rates

Open Rate_Comparison, WeeklyEvents, Pareto and DMAIC in the same workbook. This eight-week event log is separate from the 120-test investigation.

10 min · Prompt a short summary

Use DMAIC1 to review the prepared DMAIC table and rates. Independently check 93/1008=9.22619% and 11/336=3.27381%. Use the supplied Pareto chart; do not build a new chart or a full Word report.

6 min · State a decision boundary

Save three cells: measured change, limitation and next evidence. The absolute change is 5.95238 percentage points. Two concurrent changes and a short pilot do not prove a cause or sustained improvement.

Optional after class — not scheduled

Starter.xlsx and the complete prompts.md bank support rebuilding the analysis, chart and extended review later. Only the selected core prompt is required in class.

Plain-language glossary

Prompt: An instruction or question you give to an AI assistant.

Baseline: The earlier set of results used as the reference for comparison.

Control limit: A boundary calculated from the reference process data to help identify unusual behaviour.

Specification: A stated requirement that the result is expected to meet.

Failure mode: The way a component or process could fail.

RPN: Risk priority number: Severity × Occurrence × Detection.

Grounding: Supplying relevant information, such as a document passage, to help an AI answer a question.

Persona: A requested viewpoint used to review a question, not an independent expert.

Pilot: A limited follow-up or trial period used to collect more evidence.

Worked demonstration examples

31. A reusable instruction for the FMEA assistant

Show the instruction, then test it on just one failure mode.

Use: “Use the case records and supplied rating guidance.”

Return: “Give the failure, effect, possible cause and missing evidence.”

Check: “Keep supplied scores. Do not turn a possible cause into a fact.”

Use Trainer-Demonstrations/README.md for the full prompt. Open FMEA_Input, ScoringNote and Evidence. Test FM01 only. A normal Copilot prompt demonstrates the instruction if an agent is unavailable; identify the surface honestly.

32. What should the FMEA assistant say?

Example question: “What is still unproven about the low-delivery failure?”

Observed fact: Later tests recorded lower delivery.

Possible explanation: A restriction may have contributed.

Still needed: Evidence about the removed strainer and the effects of the two changes.

This is an illustrative response, not a recording of a live agent run. Open FM01 and the maintenance evidence. Ask learners to identify the fact, hypothesis and unanswered question.

35. A report draft that keeps an open issue visible

Use the same three headings every time: status, evidence and next action.

Status: Pilot delivery results improved.

Evidence: Strainer replacement and verification occurred together.

Next action: Complete the removed-strainer review before claiming a cause.

Read the source note in Trainer-Demonstrations/README.md. Show how a report template makes the distinction between observed improvement and proven cause visible. Ask why changing an open review to completed would be an error.

38. Check a grounded answer against a real passage

Use the supplied fictional document TR-401, revision 1.

Question: “Which measurement fields must we record?”

Answer: “Test ID, timestamp, volume in litres and duration in seconds.”

Check: Clause 1 lists those four fields. It does not approve a delivery limit.

Open Trainer-Demonstrations/TR-401-Training-Extract.pdf and read clause 1. Ask the follow-up about an approved limit: the correct answer is not specified. Demonstrate that a citation is useful only when the passage supports the answer.

Using the expanded 169-slide deck

Slides 1–48 are the one-day teaching route. Slide 49 introduces optional reference material. Slides 50–169 contain 30 worked examples; they are not additional scheduled labs. Keep the original six-module timetable and the 25/20/20-minute guided Excel exercises.

Reference index: SPC: slides 50–81; FMEA: slides 82–105; DMAIC: slides 106–129; Prompting: slides 130–149; Agents: slides 150–157; Grounding: slides 158–165; Security: slides 166–169

Each reference example follows data, prompt, illustrative answer and verification. Supporting CSVs, prompts and Worked-Examples.pdf are in Reference-Examples. The trainer demonstrations use the supplied Trainer-Demonstrations pack.